Document continuity — break scores

Which document each page belongs to, and the values a document's pages share.

> **Status (2026-10-19): design agreed, engine not yet rebuilt.** This file describes the break-score method the user asked for on 2026-10-19. The code still runs the gate + weighted ladder of 2026-10-18 (summarised in §13 so it can be compared). The weights below keep the fitted values of that revamp, sign-flipped so that **positive means break**; signals marked *new* are estimated and must be fitted before trusting. Background and measurement: `CONTINUITY_ENCOUNTER_REVAMP.md`.

Engine	`core-pipeline/stages/lib/continuity/engine.py`
Signals	`core-pipeline/stages/lib/continuity/signals.py`, `layout.py`, `signature.py`
Runs	at four checkpoints (§6): blank_junk pass 1 (seq 30), blank_junk pass 2 (60), page_subtype (75), `continuity` (87, final)
Settings	`stages/lib/keyword-canon/continuity_canon.json` (reloads on change)
Writes	`page_break_scores` (every checkpoint), `page_continuity_results` (final), `imaging/<chart>_continuity.csv`
Final values	written by `imaging_final` (seq 100)
Shown	review-ui: Break Before / Break After, the tag, and the basis of each score

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1. The idea

Every page gets two scores:

- **Break Before:** how strongly the evidence says a new document starts *at* this page.
- **Break After:** how strongly the evidence says the document *ends* after this page.

The tag (**New / Continue / Last / Standalone**) is worked out from the two scores. It is never stored as a fact on its own. Break After of page *i* and Break Before of page *i+1* describe the same gap, so they are the **same number**, shown from both pages. Change the evidence on one gap and the tags on both sides follow; they cannot disagree.

Nothing is final until the last checkpoint. Each checkpoint **recomputes** every score from all the evidence available at that moment (never adjusting the previous result), so a re-run gives the same answer and a later stage can undo an earlier break.

Getting it wrong is asymmetric: merging two encounters is a clinical error that reaches coding; splitting one is a nuisance a reviewer catches. Unclear gaps are **flagged**, not guessed.

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2. The gap score

```
break(i → i+1) = Σ weight × value        every value in [0, 1]; positive = break

  ending evidence   read on page i alone      (how page i ends)
+ opening evidence  read on page i+1 alone    (how page i+1 starts)
+ pair evidence     comparing the two pages   (do they belong together)
```

Gap score	Gap state
≥ **+4**	**break**
≤ **–4**	**joined**
between	**unclear**: treated as joined (the page stays in the open document, the recoverable error) and flagged `review_required`

Never widen the band to lower the unclear rate: those are the gaps most likely to be decided wrongly.

Two fixed ends: the chart's first page has Break Before = **break** and its last page has Break After = **break**. Nothing else is hard: DOS and printed page numbers are heavy weights, not gates, because OCR misreads "2 of 3" and DOS extraction makes mistakes.

2.1 Ending evidence (page i, part of its Break After)

Signal	Weight	Value	Status
`page_fill`	–5.1	how far down the body text reaches (§4.2); a full page pulls toward joined, an empty bottom adds nothing	fitted
`bottom_ink`	–3.0	C1 only, before line positions: the same measure from the image's last ink row (§4.2)	*new*
`prev_signed`	+4.4	the page is signed anywhere	fitted
`signed_in_tail`	+8.0	a signature block in the last third of the lines (replaces the old "signature terminator" override)	*new*
`page_n_of_n`	+8.0	prints "page N of N": the last page	*new*
`continued_marker`	–6.0	ends with "continued", "cont'd", "(over)", "see next page"	*new*
`cut_mid_sentence`	–2.0	the last body line ends without terminal punctuation, mid-word, or in a hyphen	*new*

2.2 Opening evidence (page i+1, part of its Break Before)

Signal	Weight	Value	Status
`cur_has_title`	+6.0	the page carries a title for its page type	estimated
`page_one`	+6.0	prints "page 1 of M"	*new*
`cur_opens_note`	–0.5	opens on Chief Complaint / HPI (fitted small and joining; kept as measured)	fitted
`starts_mid_sentence`	–2.0	the first body line starts in lowercase or mid-list	*new*

2.3 Pair evidence

Signal	Weight	Value	Status
`different_dos`	**+15.0**	both pages carry their own DOS and From or To differs	*new* (was gate 4)
`same_dos`	**−10.0**	this page's own DOS equals the open document's	*new* (was gate 3)
`pagination_follows`	**−10.0**	"3 of 5" → "4 of 5"	*new* (was gate 5)
`section_forward`	−10.0	sections move forward in note order (§4.3)	fitted
`page_type_change`	+7.0	(1 − same-type probability) × confidence in the new type (§4.4)	estimated
`same_visit_number`	−5.4	the same visit / encounter / FIN / CSN number on both	fitted
`header_line_repeats`	−4.7	a header line (≥ 12 characters) printed on both	fitted
`body_cosine_char`	−3.7	character tf-idf similarity, tail → head (§4.1)	fitted
`full_path_same`	−3.0	the same page type at every level	estimated
`footer_line_repeats`	−2.8	a footer line printed on both	fitted
`body_cosine_word`	−1.0	word tf-idf similarity	fitted
`kind_change`	+8.0	one page is Main and the other Junk; value = the junk verdict's confidence (§3)	*new*
`same_junk_group`	−4.0	both Junk with the same junk group (invoice → invoice)	*new*
`blank_between`	+1.0	one or more blank pages sit in the gap (§3.2)	*new*

`different_dos` (+15) outweighs `pagination_follows` (−10) as before: a different date is a different encounter whatever the footer says (+5 → break). Same DOS plus page numbers following on is −20, a very strong join.

`fitted` = measured on 170 labelled pairs from 15 charts (balanced accuracy 0.79 ± 0.19, split by chart); the `measurement` block in the canon carries the provenance.

2.4 The basis

Every score is stored with its parts, so a reviewer sees *why*:

```
Break Before  +11.7   break    page_type_change +5.7, cur_has_title +6.0, page_fill(prev) −0.0
Break After    −0.9   unclear  page_fill −5.1, page_type_change +4.2  → review
```

The top four contributions by size go into `evidence`; all of them are in the JSON.

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3. Main, junk and blank pages

3.1 Junk pages form their own segments

Junk pages are no longer dropped from continuity. They are scored with the same method and grouped into **junk segments** that get the same four tags: a three-page records request is `New (Junk J1)`, `Continue (Junk J1)`, `Last (Junk J1)`.

- The step between Main and Junk is `kind_change` (+8 × junk confidence): a confident junk page after a note breaks, but a weakly called one does not cut a note in half.

- If blank_junk pass 2 reverses a junk call, the next checkpoint re-scores the gap and the note joins up again.
- DOS signals are not used between junk pages.

3.2 Blank pages are compared across

Duplex scans put blank backs inside documents. A blank page is **transparent**: the gap is scored between the page before it and the page after it, plus `blank_between` (+1.0).

- The blank page has no Break Before / After of its own and no tag.
- It is shown **inside** the document when the gap across it is joined ("Blank, in Doc 3"), and **between** documents when it is a break.

3.3 Duplicates

As before: a duplicate takes its original's document, tag and values, and adds no gap.

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4. The signals in detail

4.1 Text running from one page into the next

1. **Strip the furniture**: lines of ≥ 12 characters printed on both pages (patient banner, footer template) are removed; they are what `header_line_repeats` / `footer_line_repeats` score.
2. **Tail and head blocks**: the previous page's last body lines, walking back in groups of three lines until 20 words; the next page's first body lines, forward, likewise.
3. **TF-IDF cosine** on those blocks: character 3–5 grams within words, word 1–2 grams; `min_df` 2, sublinear tf; fitted on every block of the chart. Normalised `min(1, cosine / 0.25)` for characters, `/ 0.15` for words.

4.2 Page ending: fill and bottom space

Where the body ends: the bottom of the last line **starting above the footer band**, divided by the footer top (0.92), clipped to 1. Bottom space = 1 − fill.

Fill	Value
≤ 0.60	0: the text stopped early, no pull toward joined
0.60 – 0.92	graded: (fill – 0.60) / 0.32
≥ 0.92	1: the text runs on

Sources, best first:

Checkpoint	Source
C2+	Final 2 line polygons, else Final 1 words grouped into lines
C1	`bottom_ink`: the corrected page image's last row with ink above the footer band (works on handwriting, needs no OCR boxes)
none	text proxy: 1 when ≥ 28 lines or ≥ 1,800 characters, else 0; flagged when it swings a gap

Bottom space only counts together with how the next page opens: an empty bottom followed by a titled page is a break; an empty bottom followed by a page continuing the same sections (the short last page of a long note

is the common case) stays joined.

4.3 Section order

Note order (`section_order` in the canon): chief complaint → HPI → review of systems → past history → social → family → medications → allergies → vitals → physical exam → labs → imaging → problem list → assessment / diagnosis → plan → patient instructions → follow-up.

`section_forward = clip((first section of this page − last section of the previous) / steps × 4, 0, 1)`. Moving backwards counts 0.

4.4 Page type change

Previous	This page	Confidence	Same-type probability	Value	× +7
Progress Note	Lab	90%	10%	0.81	+5.7
Progress Note	Lab	60%	40%	0.36	+2.5
Progress Note	Progress Note	95%	95%	0.05	+0.3

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5. The tag

Each side of a page is **break** or **not** (joined or unclear):

Break Before	Break After	Tag
break	not	**New**
not	not	**Continue**
not	break	**Last**
break	break	**Standalone**

Shown as `New (Doc 3)`, `Continue (Junk J1)`, and so on. A page with an unclear side carries `review_required` and a "?" in review-ui (`Continue? (Doc 3)`).

Documents are numbered in page order: `Doc n` for Main segments, `Junk Jn` for junk segments.

Field	Meaning
`start_confirmed`	the New page has a title or prints "page 1"
`link_strength`	`strong` while every gap back to the New page is ≤ −10, else `weak`

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6. Checkpoints: nothing is final early

#	Runs at	Evidence it adds	Writes
C1	blank_junk pass 1 (seq 30)	Tesseract text: titles, "N of M", header/footer repeats, signature phrases, cosines; `bottom_ink`; pass-1 verdicts (Main / Junk / Blank)	`page_break_scores`
C2	blank_junk pass 2 (60)	Final 1 / Final 2 line positions (exact `page_fill`, cut / mid-sentence), section headers, KV page numbers, pass-2 verdicts	`page_break_scores`
C3	page_subtype (75)	page type, confidence, full path, canon title	`page_break_scores`
Final	`continuity` (87)	Extracted DOS, signature flags; tags fixed, Final values carried	`page_break_scores` + `page_continuity_results`

- **Every checkpoint recomputes** from scratch with the signals present; a missing signal is left out, not counted as 0 (`evidence` records which were available).
- **Each checkpoint keeps its own row**, so review-ui can show how a page's scores moved (C1 –2.1 → C2 +6.3 → Final +6.3: "the line positions showed the text stopped at 40%").
- Only the Final checkpoint feeds Final values and `imaging_final`. Earlier checkpoints are provisional; no stage reads them as decisions today.

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7. Review flags (`review_required`)

Flag	When
Unclear	Break Before or Break After fell in the band (§2)
Long run without page numbers	a document of more than **25** pages where no page prints a number — every page of it
Text-proxy fill	page fill came from the text proxy and moved the gap across a threshold
Flip at Final	the Final tag differs from C3's and the deciding signal is weak (|contribution| < 4)

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8. What a document shares: the Final values

Unchanged (`document_finals`), applied to Main documents only:

Final value	Rule
Page type	the New page's (each page keeps its own sub-type)
DOS	the New page's, on every page
DOS of a stand-alone non-visit document	every page a `default_final_dos` type → **2222-02-02**; such a page inside a visit takes the visit's DOS
Encounter type	the New page's

Junk segments carry no Final values; each junk page keeps its own junk subtype.

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9. Encounter type and the label

9.1 Visits come from the Final segments

The encounter stage (seq 90) runs after the Final checkpoint and reads only `page_continuity_results`, never an earlier checkpoint.

Segment	Encounter type
Main document (`Doc n`)	one visit: the tier evidence (LOGIC.md §8) is gathered over every page from its New page to its Last page, and the answer is stamped on all of them
Junk segment (`Junk Jn`)	none: junk is never a visit; `encounter_type` stays empty, `reason = junk`
Blank page in a document	takes its document's answer in `imaging_final` (it is not read for evidence)
Blank page between documents	none
Duplicate	its original's, as before

The tiers, negatives and confidences are unchanged. The only change is the unit: a visit is a Main segment as the break scores drew it, and a junk page can no longer be pulled into a visit or start one.

9.2 When the document itself is unsure

Encounter type inherits the document's doubt instead of hiding it:

Situation	Effect
A gap inside the document is **unclear** (§2) and the pages on each side of it would answer **different settings** on their own	`conflict = y`, `review_required`, `reason = unclear_break`; the visit's answer is kept (the joined reading) and the alternative is in the evidence
The document's `link_strength` is `weak` and the answer came from tier 3 tie-breaking only	confidence drops one bucket (0.70 → 0.60)
`continue_applied = y` on a page whose Break Before is unclear	the page is listed in the visit's evidence as "joined across an unclear break"

So a reviewer who sees `Continue? (Doc 3)` also sees whether that doubt would change the encounter type.

9.3 The label

One label vocabulary from the tag, used by `page_continuity_results`, the CSVs, `imaging_final` and review-ui:

Tag	Main	Junk	Unclear side
new	`New (Doc 3)`	`New (Junk J1)`	`New? (Doc 3)`
continue	`Continue (Doc 3)`	`Continue (Junk J1)`	`Continue? (Doc 3)`
last	`Last (Doc 3)`	`Last (Junk J1)`	`Last? (Doc 3)`
standalone	`Standalone (Doc 3)`	`Standalone (Junk J1)`	`Standalone? (Doc 3)`
blank	`Blank (in Doc 3)` / `Blank`	—	—

Replaces `First / Continue / Last / Single (Doc n)` (`position_label` in `engine.py`).

`encounter_label` stays what it is, the setting's display name from `encounter_canon.json` (`Outpatient (F2F)`, `Outpatient (Tele)`, `Inpatient`, `Home`). review-ui shows the two side by side: `New (Doc 3) · Outpatient (F2F)`. Junk pages show only their segment label.

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10. Storage

New table (add, never drop):

```
        CREATE TABLE page_break_scores (
            id               BIGSERIAL PRIMARY KEY,
            chart_id         BIGINT NOT NULL REFERENCES chart_list(id) ON DELETE CASCADE,
            page_id          BIGINT NOT NULL REFERENCES page_list(id) ON DELETE CASCADE,
            stage_name       VARCHAR(50) NOT NULL,      -- the checkpoint
            pass_no          SMALLINT    NOT NULL,
            break_before     NUMERIC(6,2),               -- NULL on blank / duplicate pages
            break_after      NUMERIC(6,2),
            tag              VARCHAR(12) CHECK (tag IS NULL OR tag IN
('new','continue','last','standalone')),
            segment_kind     VARCHAR(10) CHECK (segment_kind IS NULL OR segment_kind IN
('main','junk')),
            segment_seq      INT,                        -- Doc n / Junk Jn
            seq              INT,                        -- position within the segment
            review_required  BOOLEAN NOT NULL DEFAULT FALSE,
            evidence         JSONB NOT NULL DEFAULT '{}'::jsonb,  -- {"before": {signal:
contribution}, "after": {...}, "available": [...]}
            created_at       TIMESTAMPTZ NOT NULL DEFAULT now(),
            updated_at       TIMESTAMPTZ NOT NULL DEFAULT now(),
            UNIQUE (page_id, stage_name, pass_no)
        );
```

`page_continuity_results` gains `break_before`, `break_after`, `tag` and `segment_kind`. The existing `position` / `relation` / `decided_by` columns stay and are still written (`position`: new → first, standalone → single; `decided_by` = the largest contribution). Readers use the new columns when present.

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11. Settings (`continuity_canon.json`)

Block	Holds
`weights`	every signal in §2.1–2.3 with `side` (ending / opening / pair), `w`, `lift`, `coverage`, `status`
`decision`	`break_at` +4.0, `join_at` −4.0, `strong_join` −10.0
`pagination`, `bands`, `semantic`, `page_fill`, `section_order`, `default_final_dos`	as before
`bottom_ink`	the ink threshold and footer exclusion for C1
`markers`	`continued_marker` and closing phrases
`review_triggers`	unclear, long run 25, text-proxy fill, flip at Final
`measurement`	provenance of the weights

The loader refuses a file whose `weights` name a signal the engine does not produce, or whose band is inverted (`join_at ≥ break_at`).

12. Worked example

Page	Kind	Evidence	Break Before	Break After	Tag
1	Main	DOS 2024-03-28, title	(chart start)	–10.0 page 1 → 2 of 3	New (Doc 1)
2	Main	"page 2 of 3"	–10.0	–22.6 banner + footer repeat, full page, HPI → Plan	Continue (Doc 1)
3	Main	ends "Electronically signed by …"	–22.6	**+12.4** signed in tail, signed	Last (Doc 1)
4	Main	Lab title, text stops at 40%	+12.4	**+11.7** title on 5, page type change	Standalone (Doc 2)
5	Main	Lab after Lab, full page	+11.7	–0.9 unclear	Continue? (Doc 3), review
—	Blank	no ink	—	—	Blank, in Doc 3
6	Main	across the blank: same sections, +1 blank	–0.9	+8.8 (Junk next, 0.96)	Last (Doc 3)
7	Junk	record request, conf 0.96	+8.8	–4.7 same junk group, footer repeat	New (Junk J1)
8	Junk	record request	–4.7	+15.0 DOS 2024-06-03 on 9, kind change	Last (Junk J1)
9	Main	DOS 2024-06-03	+15.0	(chart end)	Standalone (Doc 4)

13. What it replaces (the 2026-10-18 ladder)

Then	Now
Gates decided in order (first page, blank/junk, same DOS, different DOS, pagination, page N of N)	the same facts as heavy weights; only the chart's first and last page are fixed
Score positive = continue	positive = break; same fitted weights, sign flipped
One score per page (the gap before it)	Break Before and Break After per page; the tag comes from both
Signature terminator override	`signed_in_tail` +8
A blank or junk page always ended the document; junk had no row	blank is transparent; junk forms segments; `kind_change` weighs the junk confidence
Ran once at seq 87	recomputed at four checkpoints, each kept
`single / first / continue / last`	`standalone / new / continue / last` (old column still written)
Encounter visit = a continuity document; blank / junk pages had none	visit = a Main segment; junk segments never get one; a blank page inside a document takes its answer; an unclear gap with different settings either side flags the visit (§9)
Label `First / Continue / Last / Single (Doc n)`	`New / Continue / Last / Standalone (Doc n)`, `(Junk Jn)`, `?` on an unclear side (§9.3)